

MULTIMEDIA



UNIVERSITY

STUDENT ID NO

--	--	--	--	--	--	--	--	--	--

MULTIMEDIA UNIVERSITY

FINAL EXAMINATION

TRIMESTER OCTOBER/NOVEMBER 2024
(TERM 2430)

CCP4114 – PROGRAM DESIGN
(ALL SECTIONS)

19 FEBRUARY 2025
9.00 a.m - 11.00 a.m
(2 Hours)

INSTRUCTIONS TO STUDENTS

1. This question paper consists of **FOURTEEN (14)** pages.
2. There are **TWO (2)** sections; Section A and Section B. Section A consists of **SIX (6)** questions and Section B consists of **FOUR (4)** questions. The distribution of marks for each question is given.
3. Answer **ALL** questions.
4. Please write all your answers in the **Answer Booklet** provided.

SECTION A: SHORT ANSWERS QUESTIONS (6 QUESTIONS, 50 MARKS)**Question 1 [5 marks]**

Table 1 shows the common flowchart symbols and their descriptions.

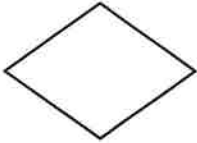
Symbols	Description
X	Indicates the beginning and end of a program
Diamond 	Z
Parallelogram 	Indicate either an input or output operation
Y	Indicates a process to be carried out (e.g: variable declaration, formula calculation)

Table 1

a) Name and draw the symbols for X and Y.

[2 marks]

b) Provide suitable description for Z.

[1 mark]

c) Draw the symbols for the on-page connector and off page connector.

[2 marks]

Continued

Question 2 [10 marks]

- a) List **four (4)** reserved words in C programming. [4 marks]
- b) List **three (3)** data types available in C programming. [3 marks]
- c) Write **two (2)** ways to define **PI** as a **CONSTANT** with the value of **3.14** in C programming. [2 marks]
- d) A repetition statement specifies that a statement or group of statements is to be executed repeatedly while some conditions remain true. [1 mark]

Question 3 [5 marks]

State whether each of the following statements is **TRUE** or **FALSE**:

- a) All variables must be defined before they are used. [1 mark]
- b) In C programming, `number` and `Number` are considered identical. [1 mark]
- c) In a flowchart, flowlines indicate the actions to be performed. [1 mark]
- d) The expression $(x > y \ \&\& \ a < b)$ is true if $x > y$ is true and $a < b$ is true. [1 mark]
- e) To indicate that 10 locations are reserved for the integer array `num`, we should write:
`num[10].` [1 mark]

Continued

Question 4 [5 marks]

Write a single coding statement that performs each of the following tasks:

- a) Define a variable `numPtr` to be a pointer to an object of type `float`. **[1 mark]**
- b) Print 1234 right justified in a 10-digit field. **[1 mark]**
- c) Write a statement that open a file `"student.txt"` for reading and assign the returned file pointer to `stdPtr`. **[1 mark]**
- d) Define variable `sum` and `x` to be of type `int`. **[1 mark]**
- e) Print `"The sum is :"` followed by a value of variable `sum`.
Assumed `sum` is declared as an integer. **[1 mark]**

Continued

Question 5 [10 marks]

Identify and correct the errors in each of the following statements. Each statement may contain one or more errors.

a) `scanf ("%d %d", &number1, number2);`

[1 mark]

b) `if (c ==> 7);`
`printf ("C is equal to or more than 7 \n");`

[2 marks]

c) `While (x <= 100)`
`total += x;`
`printf ("%d \n", &total);`
`++x;`

[3 mark]

d) `#define SIZE 10;`

[1 mark]

e) `int a [2][2] = {{1,2}, {3,4}};`
`a [1,1] = 5;`

[1 mark]

f) `void product (void)`
`{`
`int a,b,c, result;`
`fprintf ("Enter three integers \n");`
`scanf ("%d%d%d", &a, &b, &c);`
`result = a * b * c;`
`printf ("Result is %d", result);`
`return result;`
`}`

[2 marks]

Continued

Question 6 [15 marks]

Predict the output for the following statements:

a)

```
#include <stdio.h>

int main ( )
{
    printf ("%s \n","This is a string");
    return 0;
}
```

[1 mark]

b)

```
#include <stdio.h>
#define SIZE 3

int main ( )
{
    int table [SIZE][SIZE] =
        {{1,8},{2,4,6},{5}};

    int x,y;

    printf("This is the output: \n");

    for (x=0; x<SIZE; x++)
    {
        for (y=0; y<SIZE; y++)
        {
            printf("table[%d][%d] =
                %d\n", x,y,table[x][y]);
        }
    }
    return 0;
}
```

[5 marks]

Continued

c)

```
#include <stdio.h>

int main ( )
{
    int m , n;

    m = 2 % 2 + 2 * 2 - 2 / 2;
    n = ( 3 * 9 * ( 3 + ( 9 * 3 / 3 ) ) );

    printf ("The value of m is %d",m);
    printf ("\nThe value of n is %d",n);

    return 0;
}
```

[4 marks]

d)

```
#include <stdio.h>

int calCube (int n);

int main ( )
{
    int num = 5;

    printf ("Original value: %d\n",num);
    num = calCube (num);
    printf ("New value: %d\n",num);

    return 0;
}

int calCube (int m)
{
    return m * m * m;
}
```

[2 marks]**Continued**

e)

```
#include <stdio.h>
#include <string.h>

struct Student
{
    char name[50];
    char program[50];
    int stud_id;
};

int main( )
{
    struct Student student1;
    struct Student student2;

    strcpy(student1.name, "Mohaimin");
    strcpy(student1.program, "Diploma");
    student1.stud_id = 1124321;

    strcpy(student2.name, "Puteri Alia");
    strcpy(student2.program, "Diploma");
    student2.stud_id = 1123542;

    printf("Student 1 name : %s\n",
        student1.name);
    printf("Student 1 program : %s\n",
        student1.program);
    printf("Student 1 id : %d\n\n",
        student1.stud_id);

    printf("Student 2 name : %s\n",
        student2.name);
    printf("Student 2 program : %s\n",
        student2.program);
    printf("Student 2 id : %d\n",
        student2.stud_id);

    return 0;
}
```

[3 marks]**Continued**

SECTION B: DESIGNING AND PROGRAMMING (4 QUESTIONS, 50 MARKS)**Question 1 [10 marks]**

Write a complete C program containing `if...else` statements to compare **two (2) integer numbers** input by the user. The program will:

- i) Compare the numbers to determine if they are equal or not.
- ii) If the numbers are not equal, the program will indicate the relationship, specifying whether the first number is higher or lower than the second number.

Figure 1 (a), (b), and (c) show the three samples execution outputs of the program.

```
Enter first number : 5
Enter second number : 5

First number is 5
Second number is 5
First number is equal to second number.
```

Figure 1(a): Output for first number is equal to second number.

```
Enter first number : 7
Enter second number : 3

First number is 7
Second number is 3
First number is higher than second number.
```

Figure 1(b): Output for first number is higher than second number.

```
Enter first number : 9
Enter second number : 12

First number is 9
Second number is 12
First number is lower than second number.
```

Figure 1(c): Output for first number is lower than second number.

[10 marks]

Continued

Question 2 [10 marks]

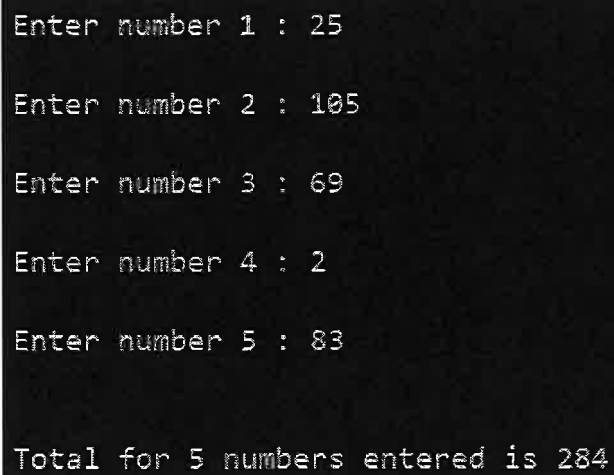
A program is designed to calculate the sum of five integers. The program should read one value each time `scanf` is executed. Use `while` loop to complete the program.

(a) Draw a flowchart to design a C program explained above.

[5 marks]

(b) Write a complete C program as explained above.

Figure 2 shows the sample output of the program.



```
Enter number 1 : 25
Enter number 2 : 105
Enter number 3 : 69
Enter number 4 : 2
Enter number 5 : 83

Total for 5 numbers entered is 284
```

Figure 2: Output for the program.

[5 marks]

Continued

Question 3 [17 marks]

Table 2 shows the employees' salary code and amount based on their positions.

Position	Salary code	Salary amount
Manager	01	RM 300 per day
Hourly worker	02	RM 25 per hour for the first 40 hours + 1.5 times hourly salary for overtime hours
Commission worker	03	RM 250 per day + 5.7% of gross weekly sale
Piece worker	04	RM 300 per week + number of items completed (RM 3 per item)

Table 2

Using switch...case and if...else statements, write a complete C program to compute the weekly pay for each employee based on the salary codes and the amounts mentioned above. Assume that all workers work five (5) days per week.

Within the program, prompt the user to enter the salary code and the necessary information required to calculate the salary.

Figure 3 (a), (b), (c) and (d) show the samples execution outputs of the program.

```
-----  
WEEKLY SALARY CALCULATOR  
-----  
Enter salary code : 01  
  
Position : Manager  
Weekly salary : RM 300.00
```

Figure 3(a): Output for manager.

Continued

```
-----  
WEEKLY SALARY CALCULATOR  
-----  
Enter salary code : 02  
  
Enter hours of working (hour) : 39  
  
Position : Hourly worker  
Weekly salary : RM 975.00
```

Figure 3(b): Output for hourly worker without overtime.

```
-----  
WEEKLY SALARY CALCULATOR  
-----  
Enter salary code : 02  
  
Enter hours of working (hour) : 67  
  
Position : Hourly worker  
Weekly salary : RM 2012.50
```

Figure 3(c): Output for hourly worker with overtime.

```
-----  
WEEKLY SALARY CALCULATOR  
-----  
Enter salary code : 03  
  
Enter number of items completed : 463  
  
Position : Piece worker  
Weekly salary : RM 1689.00
```

Figure 3(d): Output for piece worker.

[17 marks]

Continued

Question 4 [13 marks]

Write a program that plays the game of “Guess the Number”. The player should make a guess by selecting an integer in the range of 1 to 10. For each attempt, the program will respond with one of the following:

- i) Excellent! You guessed the number!
- ii) Too low. Try again.
- iii) Too high. Try again

If the player’s guess is incorrect, the program should loop until the player finally guesses the number correctly. Once the player guesses it right, the program should end.

The program should be structured as follows:

In the main ():

- i) Set a number to be guessed.
- ii) Prompt the user to guess the number.
- iii) Read the guessed number.
- iv) Pass the set and guessed numbers to response () function.
- v) When the user successfully guessed the number, print a message that the game has ended.

In response () function:

- i) Use while loop and if...else statements to print a suitable response to the guessed number.
- ii) If the user guessed the number correctly, back to the main () function to end the game.

Figure 4 shows the sample output of the program.

Continued

```
GUESS A NUMBER BETWEEN 1 TO 10  
  
Enter your guess : 6  
  
Attempt 1: Too low. Try again  
  
Enter your guess : 9  
  
Attempt 2: Too high. Try again  
  
Enter your guess : 4  
  
Attempt 3: Too low. Try again  
  
Enter your guess : 8  
  
Excellent! You guessed the number!  
  
GAME END
```

Figure 4: Sample output for the program.

[13 marks]

End of Page.